

Credit Card Customer Churn Analysis Using Machine Learning

Executive Summary

Customer churn is a critical challenge in the banking industry, as acquiring new customers is often more expensive than retaining existing ones. This project aims to identify customers who are likely to leave the bank by analyzing customer demographic and financial information using machine learning techniques.

Using the Churn_Modelling.csv dataset, exploratory data analysis (EDA), feature engineering, and predictive modeling were performed. Multiple classification algorithms were evaluated, including Logistic Regression and Random Forest. The Random Forest model achieved the best predictive performance with approximately 86% accuracy, enabling the identification of high-risk customers and supporting customer retention strategies.

1. Introduction

Background

Customer churn refers to the loss of customers who discontinue using a company's products or services. In the banking sector, customer churn can significantly impact profitability and long-term growth.

Banks collect large amounts of customer data that can be analyzed to understand customer behavior and predict future churn. Machine learning provides effective techniques for identifying patterns associated with customer attrition.

Problem Statement

The objective of this project is to predict whether a customer will leave the bank based on demographic, financial, and account-related attributes.

Business Objective

The bank aims to:

- Reduce customer churn.
 - Improve customer retention rates.
 - Identify high-risk customers.
 - Develop targeted retention campaigns.
 - Increase customer lifetime value.
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2. Dataset Description

The dataset contains information about bank customers and whether they exited the bank.

Dataset Information

Feature	Description
CreditScore	Customer credit score
Geography	Customer country
Gender	Male/Female
Age	Customer age
Tenure	Years with bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card ownership
IsActiveMember	Activity status
EstimatedSalary	Annual estimated salary
Exited	Target variable

Target Variable

Exited:

- 0 = Customer retained
- 1 = Customer churned

3. Methodology

The project followed a standard machine learning workflow:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Engineering
5. Model Building
6. Model Evaluation

4. Data Preprocessing

Data Cleaning

The following columns were removed because they do not contribute to predictive performance:

- RowNumber
- CustomerId
- Surname

Handling Missing Values

The dataset contained no significant missing values.

Encoding Categorical Variables

Categorical variables such as Geography and Gender were converted into numerical features using one-hot encoding.

Feature Scaling

Numerical variables were standardized using StandardScaler to improve model performance.

5. Exploratory Data Analysis

Customer Churn Distribution

Analysis revealed that approximately 20% of customers churned while 80% remained with the bank.

Insight

The dataset exhibits class imbalance, which is common in churn prediction problems.

Churn by Age

Older customers showed a higher tendency to leave the bank compared to younger customers.

Insight

Age appears to be a significant predictor of customer churn.

Churn by Geography

Customers from Germany displayed a higher churn rate than customers from France and Spain.

Insight

Regional differences may influence customer satisfaction and retention.

Churn by Customer Activity

Inactive members were significantly more likely to churn than active members.

Insight

Customer engagement plays a major role in retention.

Churn by Number of Products

Customers with fewer banking products demonstrated higher churn rates.

Insight

Cross-selling additional products may improve retention.

6. Feature Engineering

The following transformations were applied:

- One-hot encoding of categorical variables.
- Removal of irrelevant identifiers.
- Feature standardization.
- Creation of model-ready datasets.

These steps improved data quality and model effectiveness.

7. Machine Learning Models

Logistic Regression

Logistic Regression was used as the baseline classification model.

Advantages

- Simple and interpretable.
- Fast training and prediction.
- Provides probability estimates.

Performance

Accuracy achieved:

Approximately 81%

Random Forest Classifier

Random Forest was used as the primary predictive model.

Advantages

- Handles nonlinear relationships.
- Reduces overfitting through ensemble learning.
- Provides feature importance rankings.

Performance

Accuracy achieved:

Approximately 86%

Random Forest outperformed Logistic Regression across multiple evaluation metrics.

8. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision

- Recall
- F1-Score
- Confusion Matrix

Logistic Regression Results

Metric	Value
Accuracy	~81%

Random Forest Results

Metric	Value
Accuracy	~86%

Best Performing Model

Random Forest Classifier

The model provided the highest predictive performance and better identification of churning customers.

9. Feature Importance Analysis

The Random Forest model identified the following key factors influencing customer churn:

1. Age
2. Balance
3. Number of Products
4. Credit Score
5. IsActiveMember
6. Geography

Key Finding

Customer age and engagement level were among the strongest predictors of churn.

10. Business Insights

Several important patterns emerged from the analysis:

High-Risk Customers

Customers with the following characteristics were more likely to churn:

- Older age
- High account balances
- Low product usage
- Inactive membership status
- Residence in Germany

Low-Risk Customers

Customers with the following characteristics were less likely to churn:

- Active account usage
 - Multiple banking products
 - Long-term relationships with the bank
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11. Recommendations

Improve Customer Engagement

Implement personalized communication strategies through:

- Email campaigns
- Mobile notifications
- Loyalty programs

Cross-Selling Strategies

Encourage customers to adopt additional banking products such as:

- Savings accounts
- Loans
- Insurance products

Retention Programs

Target high-risk customers with:

- Special offers
- Fee waivers
- Reward programs

Predictive Monitoring

Deploy the machine learning model to continuously monitor customer churn risk and enable proactive intervention.

12. Conclusion

This project successfully developed a machine learning framework for predicting customer churn in the banking sector. Through data preprocessing, exploratory analysis, and predictive modeling, significant churn drivers were identified.

The Random Forest model achieved approximately 86% accuracy and demonstrated strong predictive capabilities. Analysis revealed that customer age, account balance, activity level, and product usage are major factors influencing churn behavior.

The findings can help banks proactively identify at-risk customers, improve retention strategies, and reduce revenue loss associated with customer attrition.

Future improvements may include implementing advanced algorithms such as XGBoost, addressing class imbalance using SMOTE, and deploying the model within a real-time customer analytics platform.